Fluid Universe Model

Core Equations and Cosmological Implications

1 Basic Hypothesis

The model is based on a single fundamental hypothesis:

Spacetime is a continuous, compressible, viscous fluid. All gravitational and cosmological phenomena are macroscopic manifestations of the flow and deformation of this fluid.

2 Fluid Dynamical Description of Spacetime

We describe spacetime using variables analogous to classical fluid mechanics:

- Spacetime density: $\rho(t,x)$
- Spacetime velocity field: u^μ
- Pressure tensor of spacetime: $P^{\mu\nu}$
- Bulk viscosity and shear viscosity coefficients: ζ, η

Under the condition of local conservation, the continuity equation reads:

$$\nabla_\mu (\rho u^\mu) = 0$$

The equation of motion for spacetime fluid takes the form:

$$\rho u^\mu \nabla_\mu u^\nu = -\nabla^\mu P^{\mu\nu} + \text{viscous terms}$$

In the low-speed, weak-field limit, these equations reduce to a modified Poisson equation for gravity, consistent with Newtonian gravity at solar system scales.

3 Cosmological Solution: Expansion Without Dark Energy

For a homogeneous and isotropic universe (FLRW background), the spacetime fluid satisfies:

$$\dot{\rho} + 3H(\rho + P) = 0$$

where $H = \dot{a}/a$ is the Hubble parameter.

By introducing a small bulk viscosity, we obtain an accelerating expansion solution naturally:

$$\ddot{a}/a = -(4\pi G/3)(\rho + 3P) + \text{viscous correction}$$

The late-time acceleration observed in supernova data can be reproduced without a cosmological constant Λ or dark energy.

4 Galactic Dynamics and Dark Matter Effects

On galactic scales, the velocity field of spacetime fluid exhibits rotational and vortical structures. The effective gravitational acceleration becomes:

$$a_{eff} = a_{Newton} + a_{vorticity}$$

This additional term naturally reproduces flat rotation curves without requiring cold dark matter particles.

5 Connection to Quantum Behavior

At small scales, spacetime fluid supports small-amplitude fluctuations and turbulent modes. These can be identified with quantum fluctuations, providing a possible classical origin of quantum phenomena.

6 Conclusion

The fluid universe model provides a unified, self-consistent framework for:

- Gravitation as fluid flow
- Cosmic expansion driven by fluid compressibility and viscosity
- Dark matter and dark energy as fluid structure effects
- A possible linkage between gravity and quantum mechanics

Further mathematical refinement and numerical fitting are underway.